Kiana Asgari

Stanford, CA | asgkiana@stanford.edu | +1 650-885-0861
Google Scholar | GitHub | LinkedIn

Education

Stanford University 2023-present
Ph.D. candidate, Management Science and Engineering Expected February 2028
Advisor: Andrea Montanari | GPA: 4.1/4.0
Dantzig-Lieberman Fellowship
M.S. in Statistics | GPA: 4.1/4.0 Summer 2026

Sharif University of Technology 2023
B.Sc. in Computer Science and Mathematics (double major)

Research Experience

Optimization landscapes and generalization *Annals of Statistics*, accepted 2026

- Developed a Kac-Rice framework to characterize local minima of high-dimensional empirical risk, covering generalized linear models and two-layer neural networks.
- Derived sharp estimation and prediction error asymptotics and characterized Hessian spectra for convex losses in the proportional regime.

World-Action Modeling for RL Post-Training Ongoing
With Mohammad Ghavamzadeh (Qualcomm AI Research) and Aida Afshar (Boston University)

- Developing RL post-training with learned transition dynamics, distillation, and learned feedback.

Research Intern, Max Planck Institute for Software Systems 2021-2022
Saarbrücken, Germany | Supervisor: Adish Singla

- Developed and released a **PyTorch benchmark** for reward shaping in sparse-reward environments, covering intrinsic-reward learning for policy gradients and self-supervised online reward shaping.

Model selection in high dimensions With Andrea Montanari | Ongoing

- Developing asymptotic theory for model selection and hyperparameter tuning, quantifying tradeoffs among training sample size, validation uncertainty, and prediction risk through leave-one-out expansions.

Selected Publications

K. Asgari, A. Montanari, and B. Saeed. "Local minima of the empirical risk in high dimension: General theorems and convex examples." *The Annals of Statistics*, accepted, 2026. arXiv:2502.01953.

Y. Alimohammadi and **K. Asgari**. "Mallows Model with Learned Distance Metrics: Sampling and Maximum Likelihood Estimation." Preprint, 2025. arXiv:2507.08108.

Technical Skills

Reinforcement learning: PPO, GRPO, actor-critic methods, reward shaping, intrinsic reward learning.
Post-training: On-policy and off-policy distillation, self-distillation, LoRA.
Frameworks: PyTorch, JAX, Hugging Face Transformers, vLLM, Stable-Baselines3.
Programming: Python, C++, Julia, Bash.
Evaluation: Model selection, hyperparameter tuning, cross-validation, reproducible benchmarking.

Teaching and Leadership

Head Ph.D. Mentor, Stanford DataSURF Summer 2026
Course Assistant, Stanford: MS&E 111X, 211, 220.
Teaching Assistant, Sharif: Reinforcement Learning; Convex Optimization; Applied Linear Algebra.